GENAI AND SOCIAL SCIENCE CONFERENCE

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AI-generated language remains systematically distinguishable from human communication

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MOTIVATION

LLMs increasingly used in social science simulations

Behavioural proxies

Stand in for individual people, for example as survey respondents or debate partners

Flamino et al. 2024 · Park et al. 2024

Simulations

Populations of LLM agents that post, reply and interact, for example to test news feeds

Park et al. 2023 · Törnberg et al. 2023 · Liu et al. 2025

Both rest on one assumption: **that LLMs convincingly reproduce human communication**.

Testing it is the central open problem: validation against human data (Grossmann et al. 2023; Bail 2024; Larooij and Törnberg 2025).

THE GAP

Some judges set too low of a bar for validation

Human judges

Crowd-workers or experts judge whether a text “sounds human”. Subjective, hard to scale, and a low bar: people miss subtle linguistic and emotional cues.

Park et al. 2023 · Bouleimen et al. 2026 · Clark et al. 2021

LLM judges

LLMs rate realism and coherence, yet they are poorly calibrated and favour their own outputs.

Zhou et al. 2024 · Bhattacharjee and Liu 2024 · Wang et al. 2025

ML detectors

Classifiers reliably tell AI text from human text, best where humans are fooled. Built to flag AI text, they say little about where it differs.

Ippolito et al. 2020 · Sadiq et al. 2023 · Dugan et al. 2024

Our contribution: We ask **where** AI replies differ from human ones, and **what changes** after calibration, in social media simulations.

OUR FRAMEWORK

CHAT: a Conversational Human Authenticity Test

A computational counterpart of the Turing test (Turing 1950). The model replies to the same message a real user replied to, and we compare the two replies in three ways.

1 · DETECTABILITY

Can a classifier tell them apart?

BERT classifier on reply text only. 50% means indistinguishable.

2 · SEMANTIC FIDELITY

Does it say what the human said?

Cosine similarity of sentence embeddings to the real reply.

3 · INTERPRETABLE FEATURES

Where exactly does it differ?

Random forest on style features; Empath topic and affect categories (Fast et al. 2016).

BENCHMARK

Three platforms, nine open-weight models

Real conversations

Platform	Test users (20 replies each)
Twitter/X	275 users
Reddit, r/politics	492 users
Bluesky	59 user

Users split into training and held-out test sets.

Models, 4 to 8 billion parameters, plus one 70B

Base: Apertus-8B, DeepSeek-R1-Distill-8B, Llama-3.1-8B, Llama-3.1-70B, Mistral-7B

Instruction-tuned: Gemma-3-4B, Llama-3.1-8B-Instruct, Mistral-7B-Instruct, Qwen-2.5-7B-Instruct

Default sampling: T = 0.8, top-p = 0.9, top-k = 50.

THE TASK

Reply to a real post, as a real user

PROMPT · REFERENCE CONFIGURATION (BASELINE + PERSONA)

PERSONA · from the user's past posts

You are @user, a politically engaged individual with progressive views. You communicate in a direct, sometimes passionate tone, often sharing news articles and commentary on current events. …

CONVERSATION

Here is a conversation in which you, as @user, are participating:
[a real post from Twitter/X, Reddit or Bluesky]

INSTRUCTION

Continue the conversation naturally by adding @user's reply of a length which is appropriate for Twitter/X.

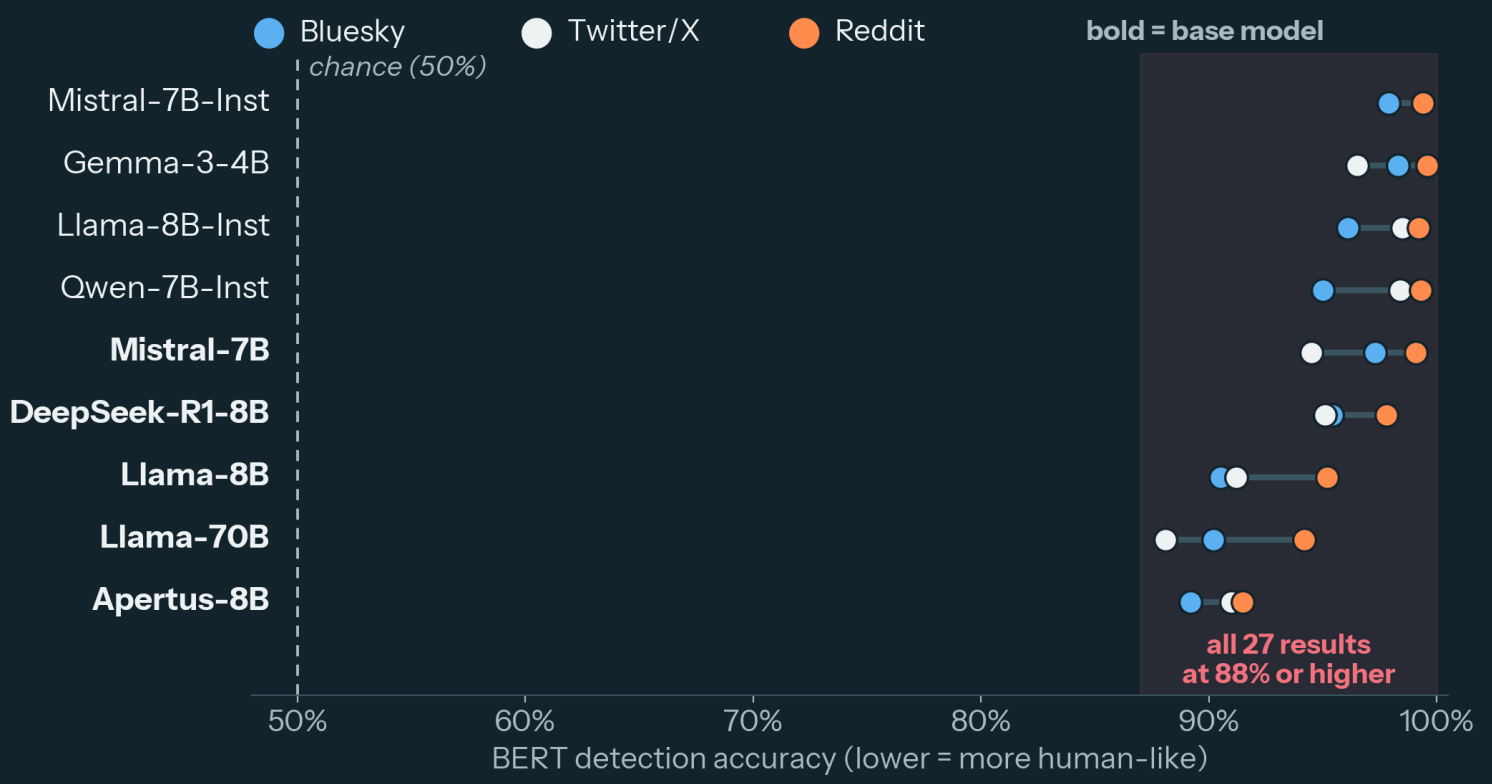
AI reply
written by each of the nine models

compared with

Real reply
what the same user actually wrote, never shown to the model

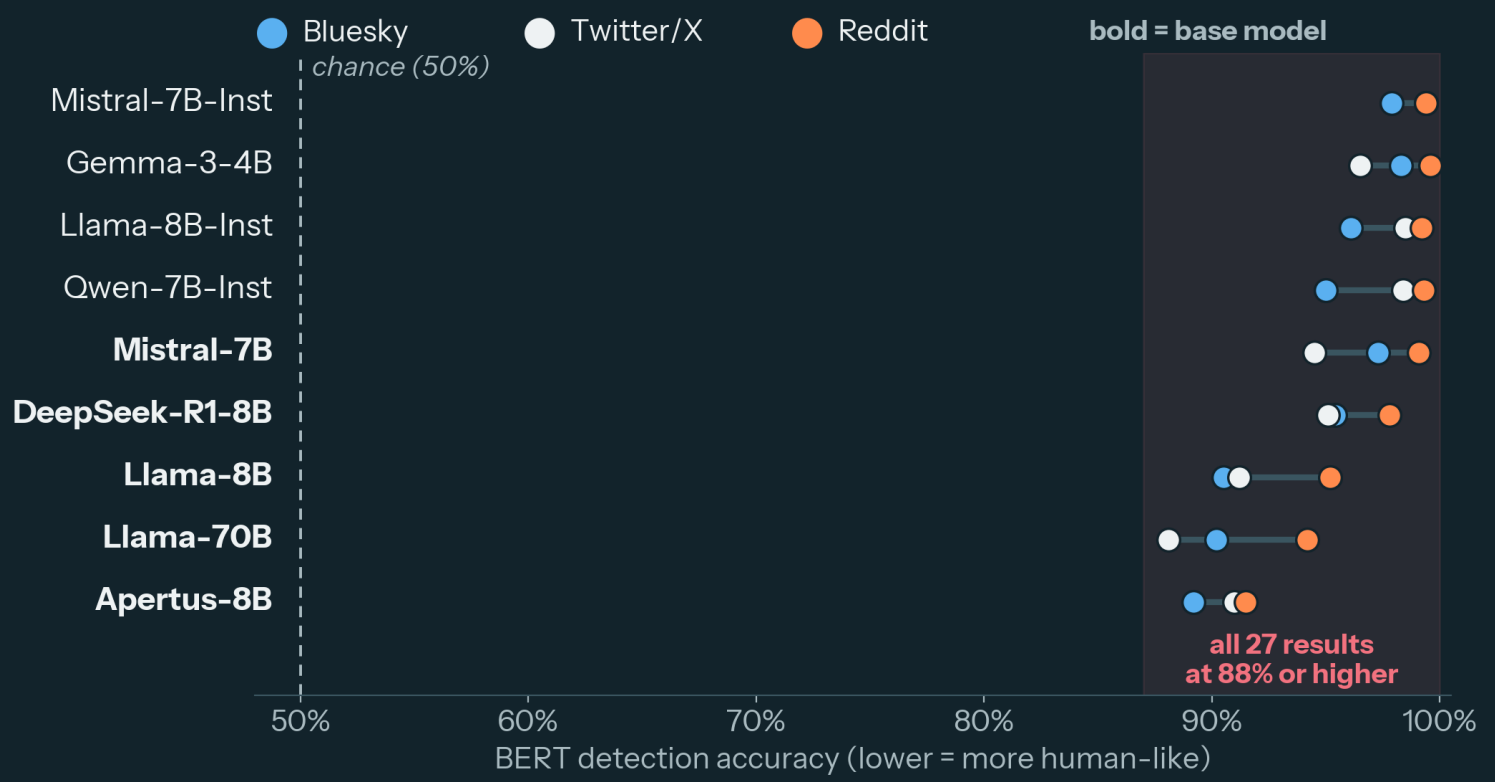
RESULT 1 · DETECTABILITY

Classifiers spot the AI reply almost every time



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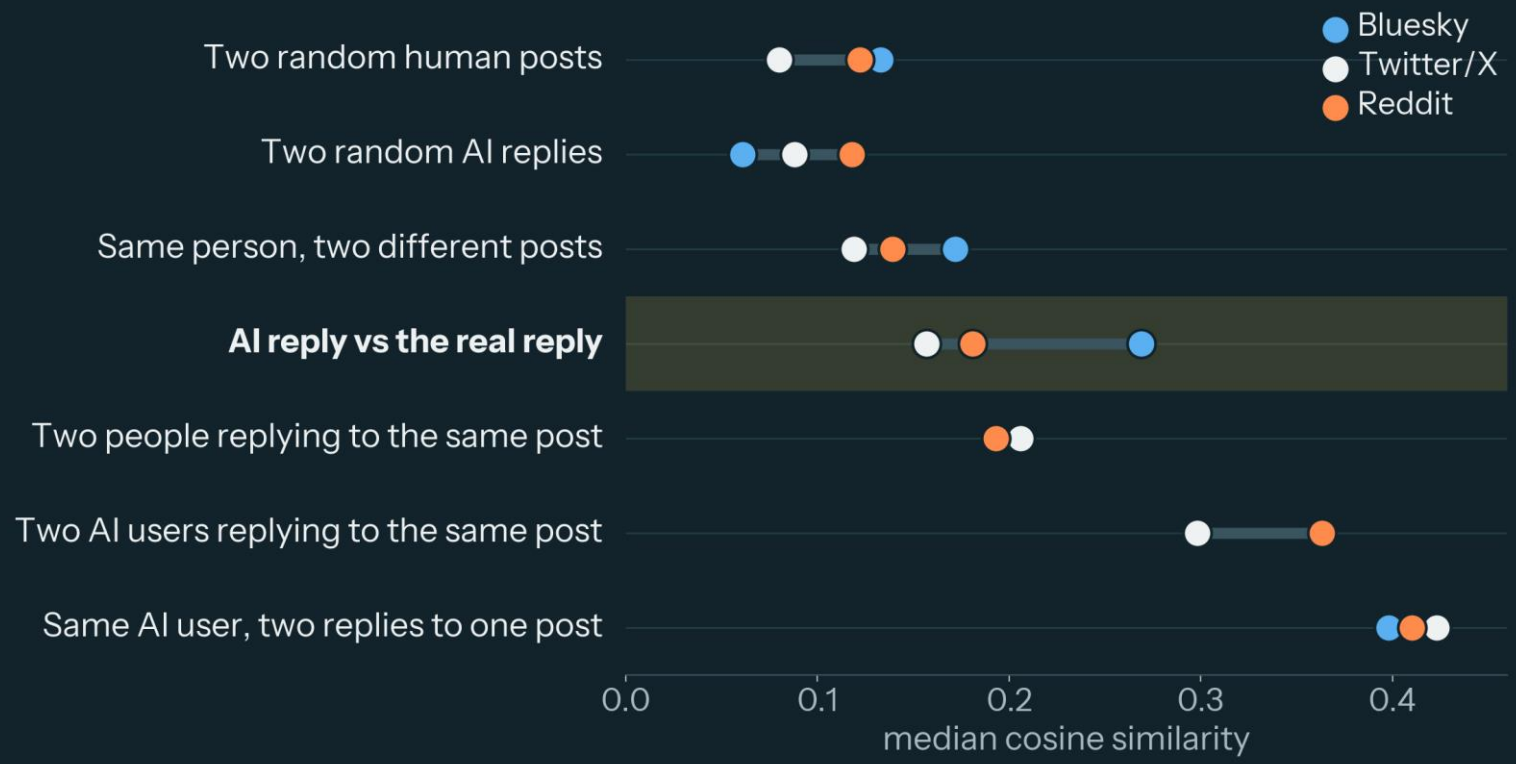
Accuracy > 88%. Reddit is the hardest platform to imitate.

Base models mimic best, instruction-tuned ones worst.

Scale does not help: Llama-70B is on par with Llama-8B.

RESULT 2 · SEMANTIC FIDELITY

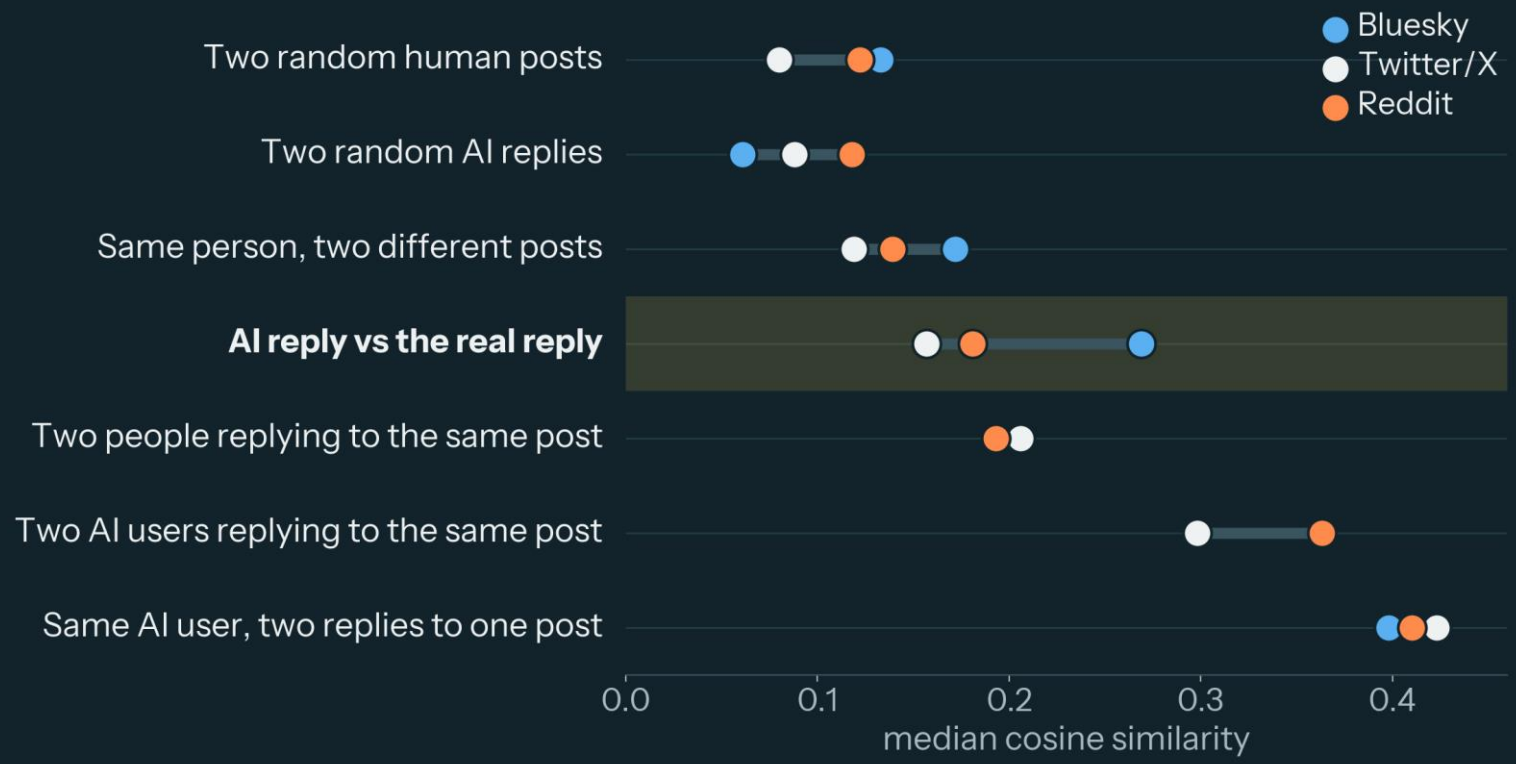
AI replies stay on topic, and resemble each other



Each row compares two texts in embedding space.

RESULT 2 · SEMANTIC FIDELITY

AI replies stay on topic, and resemble each other



Each row compares two texts in embedding space.

The AI reply is closer to the real reply than random pairs or the same person's other posts.

Two people answering the same post are closer still.

AI replies to one post are the most alike: models compress the variety of human answers (see also Wang et al. 2025).

RESULT 3 · WHAT GIVES AI AWAY: STYLE

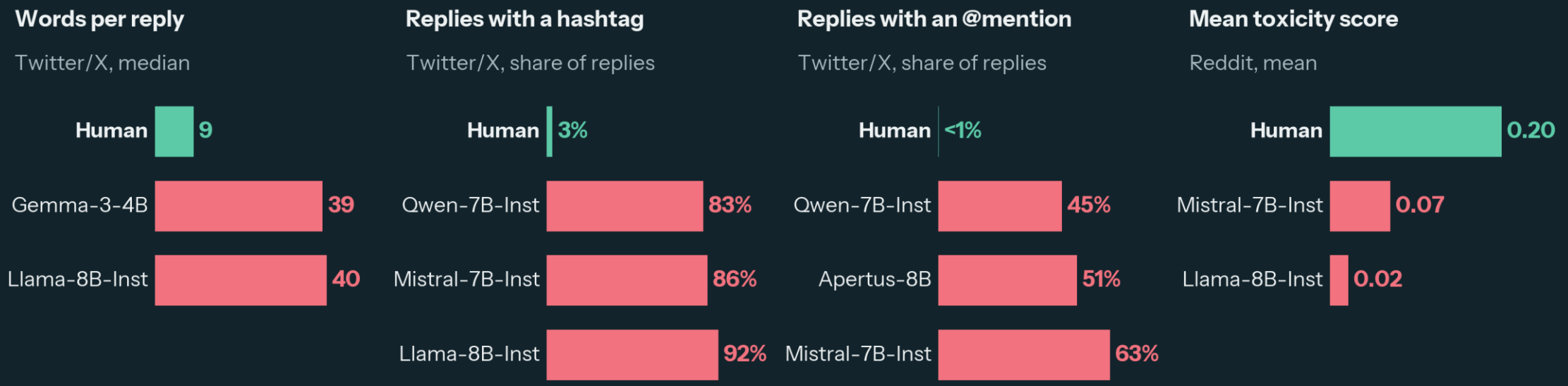
AI replies are longer, politer, and full of hashtags

Across all models, toxicity, word count and word length rank among the top classifier features on every platform.

RESULT 3 · WHAT GIVES AI AWAY: STYLE

AI replies are longer, politer, and full of hashtags

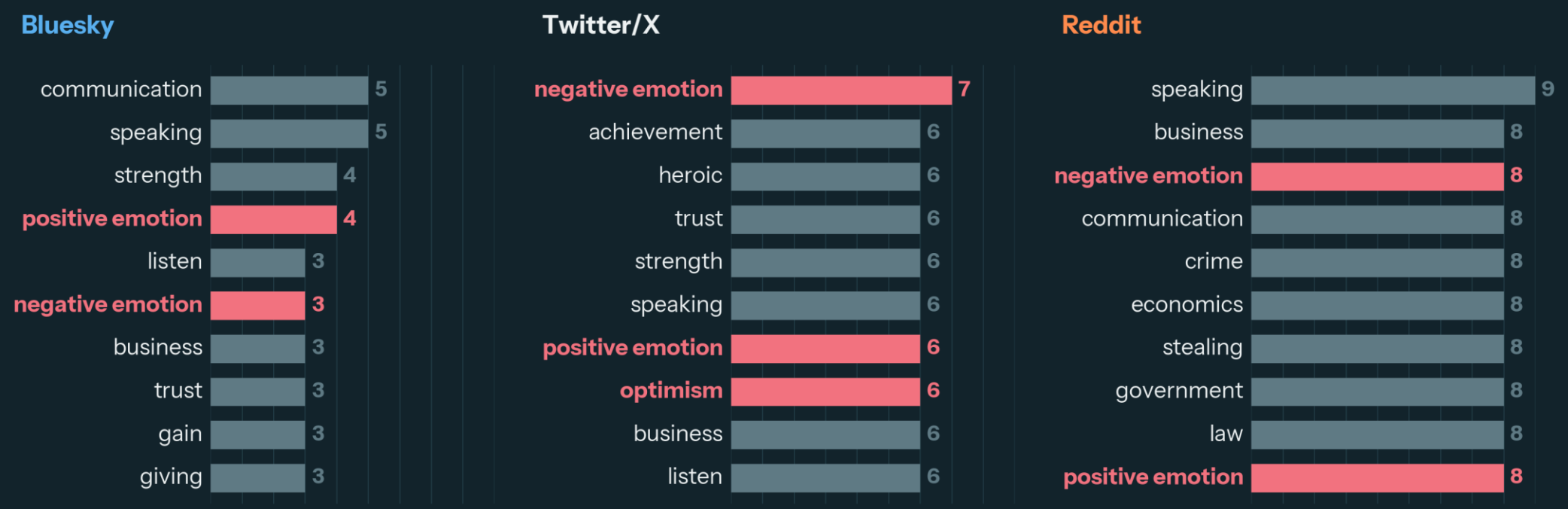
Across all models, toxicity, word count and word length rank among the top classifier features on every platform.



Selected examples. Instruction-tuned models often fail to mimic content because of the number of words, or use of hashtags, mentions, or because of toxic content.

Base models fail in their own way: Llama-8B and Mistral-7B often produce near-empty replies.

Topics and emotion differ for most models



Models (of 9) that differ significantly from humans on each of the 10 most divergent Empath categories. **Red marks emotion.**

CAN CALIBRATION CLOSE THE GAP?

Five configurations, each adding one step

BL

Baseline prompt: reply to this message

+ PE

Persona of the user, written by an LLM (reference)

+ SE

10 stylistic examples of the user's past replies

+ CR

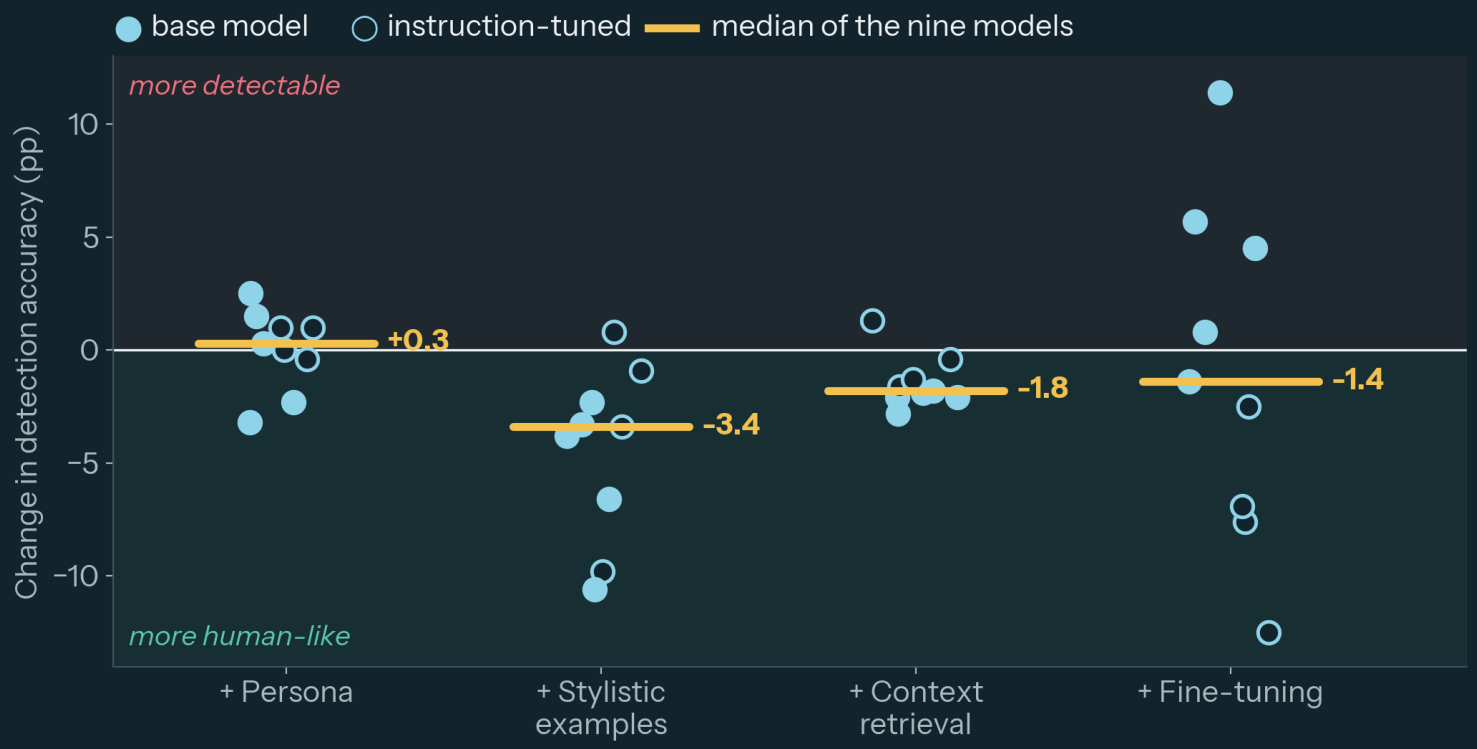
Context retrieved from the user's prior posts

+ FT

Fine-tuning (LoRA) on the users' own messages

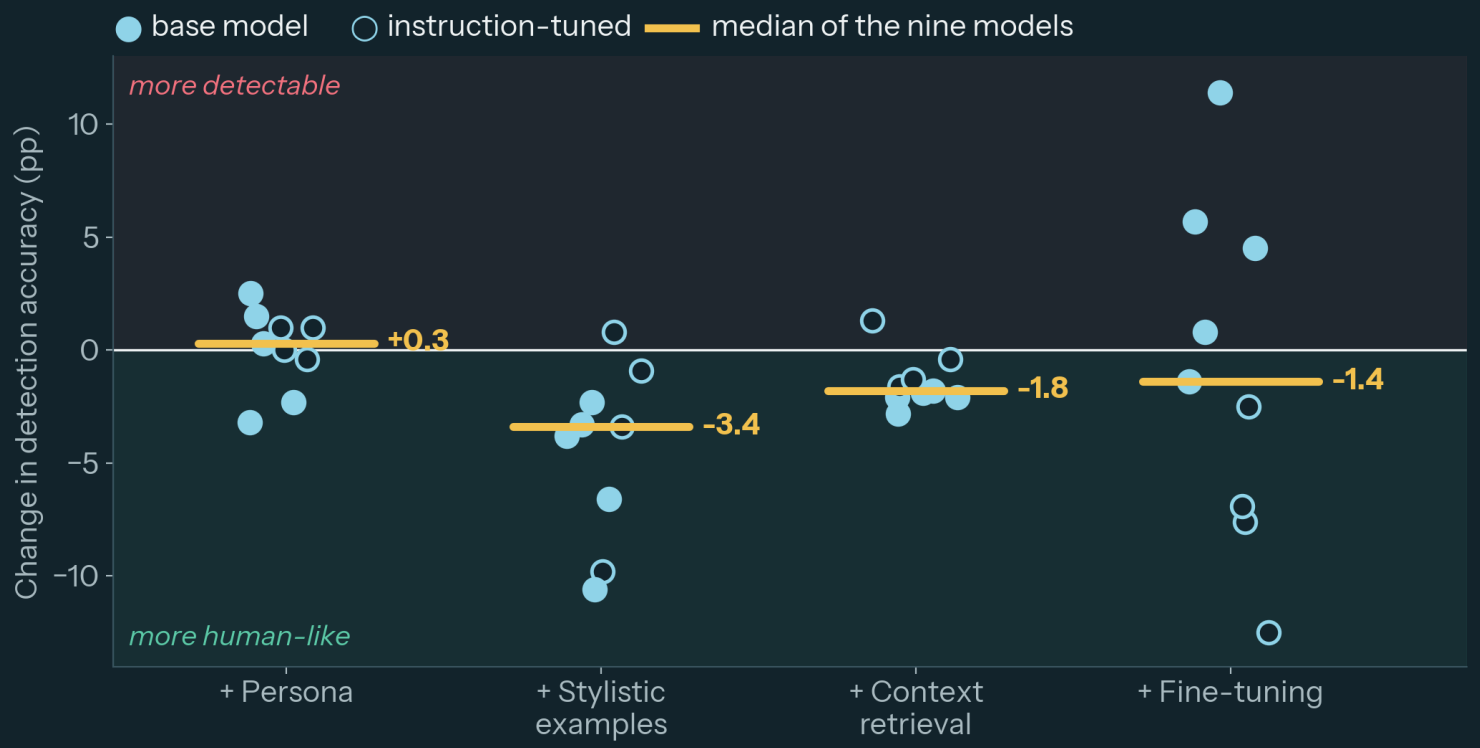
RESULT 4 · WHICH STEP HELPS

Concrete examples and context help the most



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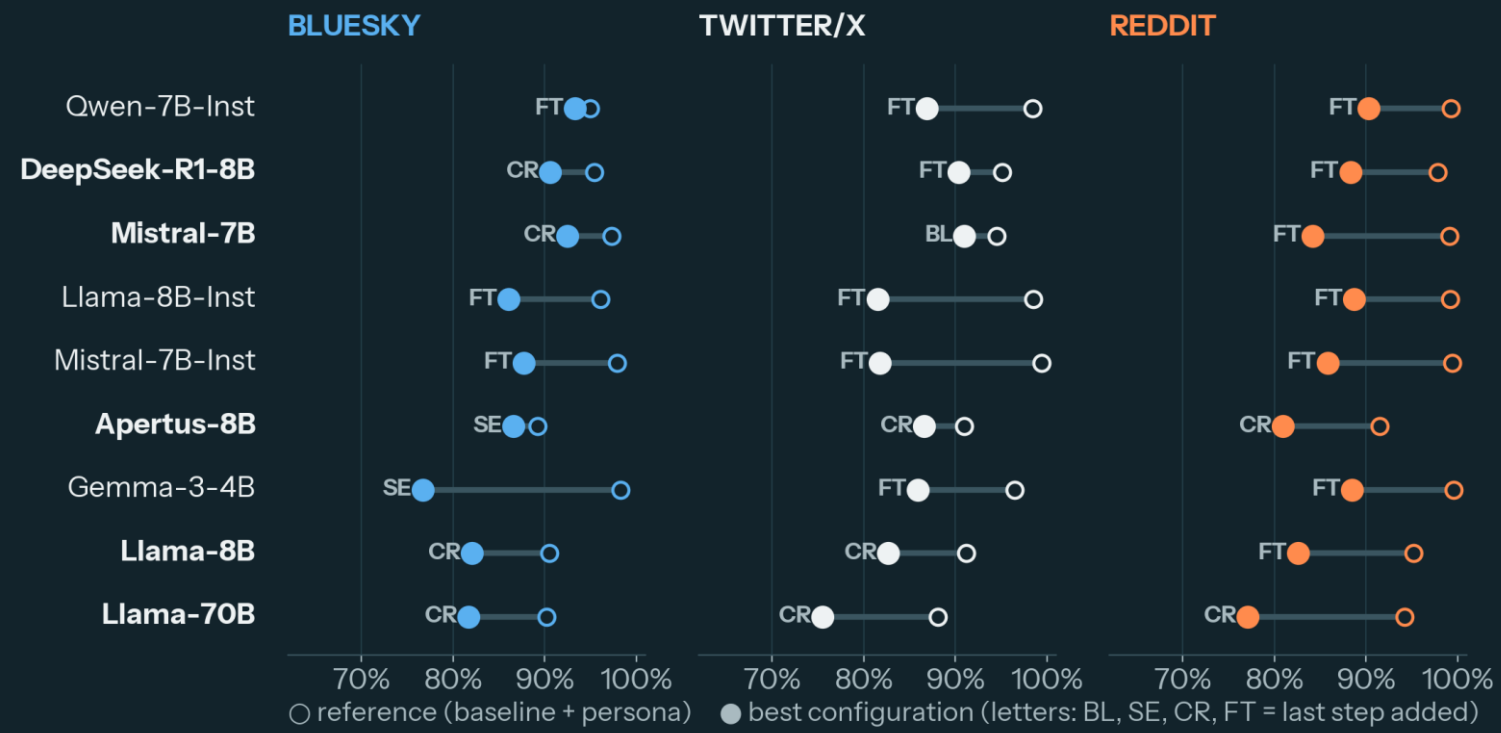
Persona: no gain, sometimes worse.

Examples and context: help almost every model.

Fine-tuning: helps all four instruction-tuned models, makes four of the five base models more detectable.

RESULT 5 · BEST CALIBRATION

Calibration buys about 10 points



Mean best accuracy: **86%** Bluesky, **85%** Twitter/X, **85%** Reddit.

Llama and Apertus base models reach **75 to 87%**; the lowest is Llama-70B on Twitter/X at 75.5%.

Models end up closer to each other.

CAN CALIBRATION CLOSE THE GAP?

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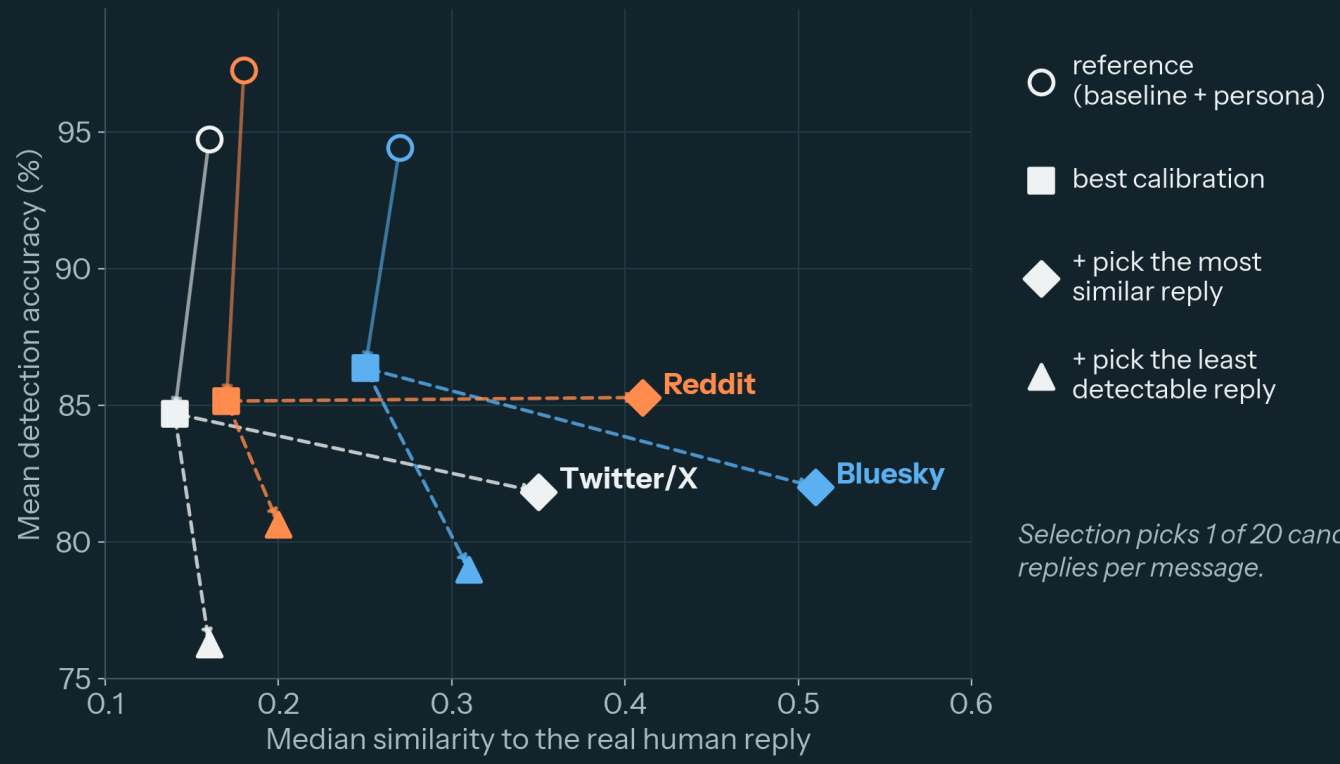
Fine-tuning (LoRA) on the users' own messages

THEN · POST-GENERATION SELECTION

Generate 20 candidate replies and keep either the first generated one, or the one **closest in meaning** to the real reply, or the one an ML classifier finds **least detectable**.

RESULT 6 · THE TRADE-OFF

Sounding human and meaning the same pull apart



reference (baseline + persona)

best calibration

+ pick the most similar reply

+ pick the least detectable reply

Selection picks 1 of 20 candidate replies per message.

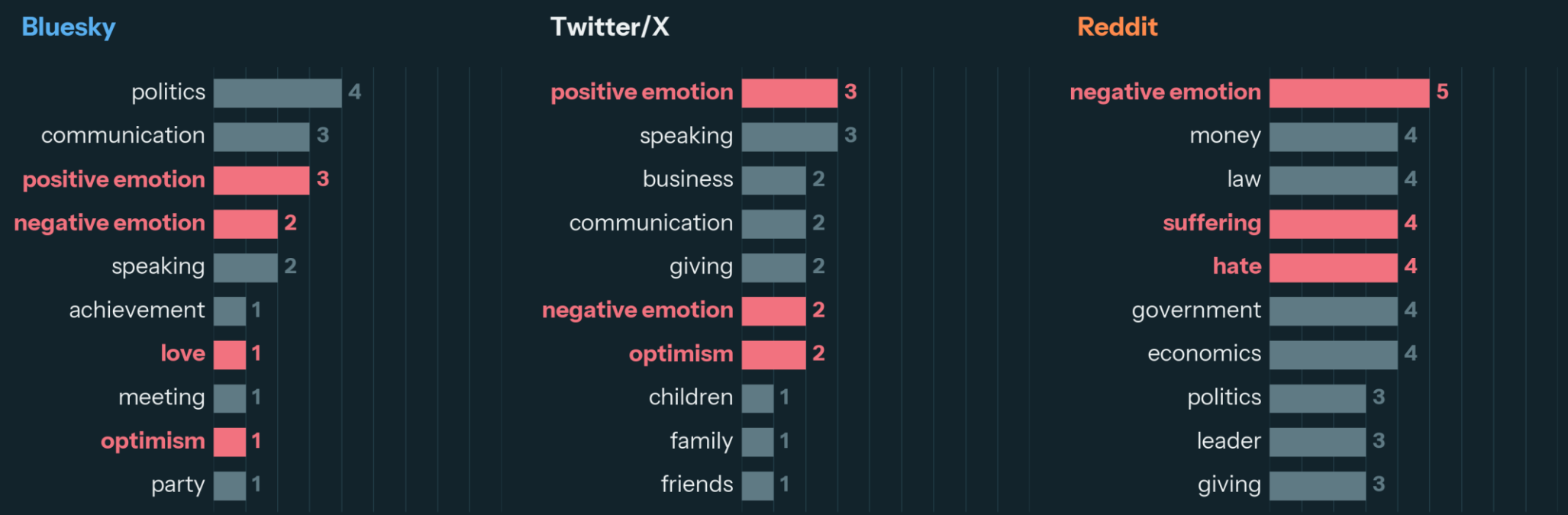
Picking for **meaning** doubles similarity; detection drops only slightly.

Picking for **human-likeness** cuts detection by another 5 to 10 points, at reference-level similarity.

The two picks overlap in only 5 to 10% of cases.

RESULT 7 · WHAT REMAINS

After calibration, topics improve and emotion remains



On each category, **fewer models differ** than before calibration. Emotion tops the list on Twitter/X and Reddit. **Red marks emotion.**

TAKE-AWAYS

1

Still distinguishable

After the best calibration and selection, classifiers still detect AI replies with 70 to 80% accuracy.

2

Examples help, personas add little

Real past replies and retrieved context work best. Base models mimic better than instruction-tuned ones; scale does not help.

3

Realism and meaning trade off

What makes a reply sound human moves it away from what the human actually said.

4

Emotion is the hardest part

Models reproduce the form of online talk better than its feeling. Validate before using them as proxies.

Thank you

Paper

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BACKUP · THE FULL PROMPT

Each calibration step adds a block to the prompt

PROMPT · ALL STEPS (PE + SE + CR)

PERSONA (PE)
You are @user, a politically engaged individual with progressive views. …

+ STYLE EXAMPLES (SE)
These are some posts that represent how you, @user, write: [10 real past posts]

+ RETRIEVED CONTEXT (CR)
This is some useful context retrieved from @user's history: [relevant past posts]

CONVERSATION
Here is a conversation in which you, as @user, are participating: [the real post]

INSTRUCTION
Continue the conversation naturally by adding @user's reply of a length which is appropriate for Twitter/X.

AI reply
written by each of the nine models

compared with

Real reply
what the same user actually wrote, never shown to the model

CHAT: detectability, meaning, interpretable features

+ FT: the model is also fine-tuned (LoRA) on the users' own messages. Then: pick 1 of 20 candidates.

Detection and similarity per platform

Configuration	Bluesky	Twitter/X	Reddit	Bluesky	Twitter/X	Reddit
Reference (BL + PE)	94.4%	94.7%	97.3%	0.27	0.16	0.18
Best calibration	86.4%	84.7%	85.2%	0.25	0.14	0.17
+ most similar of 20	82.0%	81.8%	85.3%	0.51	0.35	0.41
+ least detectable of 20	79.0%	76.3%	80.7%	0.31	0.16	0.20

Left: BERT accuracy, mean over the nine models (Fig. 10). Right: median cosine similarity to the human reply over all models (text and Fig. 11; the last row is read from Fig. 11). Most remaining errors are human replies flagged as AI (SI section 13).

BACKUP · SCOPE AND LIMITS

What the benchmark does and does not show

One human reply per post

Many replies could fit a post, so similarity serves as a reference point.

Descriptive benchmark

All configurations use the same pool of reply pairs; results compare detectability within this set-up.

Data quirks

Bluesky “users” are clusters of similar accounts. The Twitter/X same-post comparison rests on 18 pairs.

A first requirement

Matching the surface statistics of human text gives no guarantee of human attitudes or reasoning.

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